# The Algebraic Fundamental Group

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## 1 The Algebraic Fundamental Group

**Proposition 1.** *Let $k$ be a perfect field, $X$ an integral proper $k$-curve with function field $K$, and $U \subseteq X$ a non-empty open subset. Fix $K_s$ a separable closure of $K$. The composite $K_U$ of all finite subextensions $L/K$ in $K_s$ so that the corresponding finite morphism of proper normal curves is étale above all $P \in U$ is a Galois extension of $K$, and each finite subextension $K_U/K$ comes from a curve étale over $U$.*

*Proof.* It suffices to show that $K_U$ is preserved under the action of $\mathrm{Gal}(K_s/K)$ (use for instance, the criterion of étale involving the inertia group). Therefore $K_U$ is Galois over $K$.

For the last statement it suffices to show that if $L \subset K_s$ comes from such a morphism, so too does $L' \subset L$, and the compositum of two such fields coming from such a morphism also come from such a morphism. The first one comes from a previous lemma which shows that the composition of two finite separable morphisms of integral normal affine curves is étale if and only if each morphism is, and the second comes from the criterion of inertial groups, which follows from

$$\mathrm{Gal}(K_s/LM) = \mathrm{Gal}(K_s/L) \cap \mathrm{Gal}(K_s/L)$$

□

**Definition 1.** We define in this case the *algebraic fundamental group* $\pi_1(U)$ to be $\mathrm{Gal}(K_U/K)$.

As we saw with the equivalence of categories between separable extensions and transitive $\mathrm{Gal}(K)$-actions it is necessary to study étale extensions as well. The way we extend this analogy to curves is to eliminate the assumption the cover is integral.

**Definition 2.** A *proper normal curve* is a finite disjoint union of integral proper normal curves. The ring of rational functions is the direct product of the function fields of each of the components.

A finite mophism $\varphi : Y \to X$ from a proper normal curve to an integral proper normal curve is *separable* if the ring of rational functions on $Y$ is étale over the rational functions of $X$.

**Theorem 1.** *Let $X$ be an integral proper normal curve over a perfect field $k$, and let $U \subset X$ be a non-empty open. The cateogry of proper normal curves $Y$ equipped with a finite separable morphism $\varphi : Y \to X$ étale over $U$ is equivalent to the category of finite continuous left $\pi_1(U)$-sets.*

As we may functorially embed any affine normal curve into a proper one, we get the following corollary via these embeddings.

**Corollary 1.** *The category of normal affine curves $V$ equipped with a finite étale morphism $V \to U$ is equivalent to the category of finite continuous left $\pi_1(U)$-sets.*

We have thus seen how to view the fundamental group as an inverse limit of morphisms of fields coming from étale covers. In topology one is able to construct a universal cover and thereby bypass the limit. Something similar may be done in the algebraic case.

Take the integral closure $\widetilde{\mathcal{O}(U)}$ of $\mathcal{O}(U)$ in $K_U$. For any étale cover $V \to U$ we have that $\mathcal{O}(V)$ is integrally closed and so

$$\mathcal{O}(V) = \widetilde{\mathcal{O}(U)} \cap K(V)$$

Moreover, for a maximal ideal $M$ of $\widetilde{\mathcal{O}(U)}$ the intersection with $\mathcal{O}(V)$ is maximal and defines a closed point.

Now we may view $\widetilde{\mathcal{O}(U)}$ as an inverse limit as follows: note that

$$\widetilde{\mathcal{O}(U)} = \cup \mathcal{O}(V)$$

and also the set of maximal ideals of $\widetilde{\mathcal{O}(U)}$ is exactly the inverse limit of the maximal ideal of the system of $\mathcal{O}(V)$. We may then give the set of maximal ideals $\widetilde{U}$ the inverse limit topology. From this we may define a sheaf of rings in the typical way. This creates a *pro-étale cover* of $U$ which takes the rôle of the universal cover. We shall denote this by $\widetilde{X}$.

Because out discussion essentially follows the construction of the unramified covers of Riemann surfaces, we have the following understanding of the algebraic fundamental group over $\mathbb{C}$.

**Theorem 2.** *Let $X$ be an integral proper normal curve over $\mathbb{C}$, and let $U \subset X$ be an open subset. The algebraic fundamental group $\pi_1(U)$ is isomorphic to the profinite completion of the topological fundamental group of the Riemann surface associated to $U$. Hence as a profinite group it has a presentation*

$$\langle a_1, b_1, \ldots, a_g, b_g, \gamma_1, \ldots, \gamma_n : [a_1, b_1] \cdots [a_g, b_g]\gamma_1 \cdots \gamma_n = 1 \rangle$$

*where $n$ is the number of points of $X$ lying outside $U$ and $g$ is the genus of the associated Riemann surface.*

We will show in the next chapter that this result holds not just for $\mathbb{C}$ but any algebraically closed field of characteristic $0$ by the following result:

**Theorem 3.** *Let $k \subset L$ be an extension of algebraically closed fields of characteristic $0$, $X$ an integral proper normal curve over $k$, and $U \subset X$ an open subset. The base change $Y \mapsto Y_L$ induces an equivalence between finite covers of $X$ étale over $U$ and those of $X_L$ étale over $U_L$.*

*Therefore there is an isomorphism $\pi_1(U_L) \cong \pi_1(U)$.*

This theorem allows us to compute the algebraic fundamental group over one algebraically closed field in place of another.

**Corollary 2.** *Let $k$ be an algebraically closed field of characteristic 0, $X$ an integral proper normal curve over $k$, and $U \subset X$ an open subset. Then $\pi_1(U)$ has a presentation as in Theorem 2*

*Proof.* Look at an algebraic closure containing the two. □

Because of the correspondence between the algebraic fundamental group over algebraically closed fields (i.e. $\mathbb{C}$) and the profinite completion of the topological fundamental group, we have the following calculations.

**Example 1.** The following hold for any algebraically closed field:

1. $\pi_1(\mathbb{P}^1) = \{1\}$.
2. $\pi_1(\mathbb{P}^1 \setminus \{0, \infty\}) \cong \widehat{\mathbb{Z}}$.
3. $\pi_1(\mathbb{P}^1 \setminus \{0, 1, \infty\})$ is the free profinite group on two generators.

## 2 The Outer Galois Action

We now consider the algebraic fundamental group over a non-algebraically closed field and derive some interesting arithmetic results.

Let $X$ be an integral proper normal curve over a perfect field $k$. Let $K$ be the function field of $X$, and fix an algebraic closure $\overline{k}$ of $k$. Assume that $X$ is geometrically integral. Then the function field under the base change is the field $X_{\bar{k}} = K \otimes \bar{k}$.

What we will see is that quotienting the fundamental group of $X$ by that of $X_{\bar{k}}$ yields the absolute Galois group of $k$.

**Proposition 2.** *Let $X$ be a geometrically integral proper normal curve over a perfect field $k$, and $U \subset X$ an open subset. There is an exact sequence of profinite group*

$$1 \to \pi_1(U_{\bar{k}}) \to \pi_1(U) \to \mathrm{Gal}(\bar{k}/k) \to 1$$

*Proof.* We first show how to realize an element of $\pi_1(U)$ as an element of $\mathrm{Gal}(\bar{k}/k)$.

There is an isomorphism $\mathrm{Gal}(\bar{k}/k) \cong \mathrm{Gal}(K\bar{k}/K)$, so it will suffice to show that $K\bar{k} \subseteq K_U$.

Let $K_s$ be a separable closure of $K$ containing $\bar{k}$. Then we may identify $K\bar{k}$ with $K \otimes \bar{k}$. This may also be seen as the composite of all function fields of base changes $X_L$ with $L/k$ finite (and separable by perfection). Now the morphisms $X_L \to X$ are finite étale as they are base change and therefore $K\bar{k}$ is contained in $K_U$.

We now consider the kernel of this map. By considering $\pi_1(U) = \mathrm{Gal}(K_U/K\bar{k})$ and $\mathrm{Gal}(\bar{k}/k) = \mathrm{Gal}(K\bar{k}/K)$, we may peg the kernel to be

$$\mathrm{Gal}(K_U)/K\bar{k})$$

Now compare this to

$$\pi_1(U_{\bar{k}}) = \mathrm{Gal}(K_{U_{\bar{k}}}/K\bar{k})$$

and we see that we want to show we may descend a subextension of $K_{U_{\bar{k}}}$ to a subextension of $K_U$.

Take $G$ to be a finite quotient of $\pi_1(U_{\bar{k}})$. This corresponds to a finite Galois extension $K_0$ inside of $K\bar{k}$ which is the function field of a finite Galois branched cover $Y_0 \to X_{\bar{k}}$ which is étale over $U_{\bar{k}}$.

Take $f \in (K\bar{k})[t]$ for which $K_0 \cong K\bar{k}[t]/f(t)$. We then find a finite extension $L/k$ in $\bar{k}$ so the coefficients of the polynomial lie in $KL$. Then take $L_0$ to be the Galois extension of $KL$ defined by $f$. Then $L_0\bar{k} = K_0$, and $L_0$ is the function field of a finite branched cover $Y \to X_L$. Then $Y$ is étale over $U_L$. Then $Y \to X$ is étale over $U$, as the composition of two étale maps. This tells us that $L_0 \subseteq K_U$, and $L_0 \cap K\bar{k} = KL$. Therefore we can view $\mathrm{Gal}(L_0/KL)$ as a finite quotient of $\ker(\pi_1(U) \to \mathrm{Gal}(K\bar{k}/K))$. We also have $K_0 \subseteq K_U$, and therefore by varying the finite quotients of $\pi(U_{\bar{k}})$ we see that $K_{U_{\bar{k}}} \subseteq K_{U_k}$, and therefore there is a surjection $\mathrm{Gal}(K_U/K\bar{k})$ onto $\pi_1(U_{\bar{k}})$. This map is also an injection so an isomorphism as we showed for any finite quotient of $\pi_1(U_{\bar{k}})$ there is a finite quotient of $\mathrm{Gal}(K_U/K\bar{k})$ to which it maps isomorphically. [Here we use the fact this are profinite groups.] $\square$

Now from the exact sequence

$$1 \to \pi_1(U_{\bar{k}}) \to \pi_1(U_k) \to \mathrm{Gal}(\bar{k}/k) \to 1$$

we view $\pi_1(U_{\bar{k}})$ as a subgroup of $\pi_1(U_k)$, so we can view $\pi_1(U_k)$ acting via conjugation on $\pi_1(U_{\bar{k}})$. The elements of this action coming from $\pi_1(U_{\bar{k}})$ are called the *inner action*, and quotienting out by

this elements gives the *outer automorphisms*. We then identify $\mathrm{Gal}(\bar{k}/k)$ with these outer automorphisms. With this we have a continuous homomorphism

$$\rho_U : \mathrm{Gal}(\bar{k}/k) \to \mathrm{Out}(\pi_1(U_{\bar{k}})$$

We want to look at how this action acts on the "universal cover." Let $\widetilde{Q}$ be a pro-point of $\widetilde{X}$ above a point $P$ in $X$. Let $D_{\widetilde{Q}}$ be its stabilizer in $\pi_1(U)$. The residue field $\kappa(\widetilde{Q})$ is an algebraic closure of $\kappa(P)$. We then have a surjective homomorphism

$$D_{\widetilde{Q}} \to \mathrm{Gal}(\overline{\kappa(P)}/\kappa(P))$$

The kernel is called the *inertia group at* $\widetilde{Q}$. and is denoted $I_{\widetilde{Q}}$.

Note from the previous proposition we may also view $\widetilde{X}$ as being a Galois branched cover of $X_{\bar{k}}$, and the Galois group is $\pi_1(U_{\bar{k}})$.

Using this action, we may compare the stabilizers of point in $\widetilde{X}$ over a $k$-rational point under the actions of $\pi_1(U)$ and $\pi_1(U_{\bar{k}})$.

**Lemma 1.** *Let $P$ be a closed $k$-point of $X$. The stabilizer of a point $\widetilde{Q}$ of $\widetilde{X}$ lying above $P$ in $\pi_1(U_{\bar{k}})$ is the inertia group $I_{\widetilde{Q}}$ in $\pi_1(U)$.*

*Proof.* We may identify both of these as the kernel of the maps $D_{\widetilde{Q}} \to \mathrm{Gal}(\bar{k}/k)$, as the condition for the element of the stabilizer in $\pi_1(U)$ to be in $\pi_1(U_{\bar{k}})$ is to fix $\bar{k}/k$, and this is also what it means to be in the inertia group, as the extensions of residue fields of $Q$ over $P$ is $\bar{k}/k$. □

This identification of the stabilizer with the inertia group allows us to find an obstruction for such a curve to have a $k$-rational point.

**Corollary 3.** *If $U$ contains a $k$-rational point, the exact sequence*

$$1 \to \pi_1(U_{\bar{k}}) \to \pi_1(U_k) \to \mathrm{Gal}(\bar{k}/k) \to 1$$

*splits.*

*Proof.* Take $\widetilde{Q}$ in $\widetilde{U}$ lying above a $k$-rational point, and let $\overline{Q}$ be its image in $U_{\bar{k}}$. Then $\widetilde{Q}$ has trivial stabilizer in $\pi_1(U_{\bar{k}})$, so $I_{\widetilde{Q}}$ is trivial and we have an isomorphism $\pi_1(U) \to \mathrm{Gal}(\bar{k}/k)$ which gives the splitting. □

**Theorem 4** (Belyi)**.** *Let $X$ be an integral proper normal curve defined over an algebraically closed field $k$ of characteristic 0. There exists a morphism $X \to \mathbb{P}^1_k$ étale over $\mathbb{P}^1_k \setminus \{0, 1\infty\}$ if and only if $X$ can be defined over $\overline{Q}$.*

*Proof.* We just give a sketch of the proof here. The only if direction of this proof comes from the isomorphism between étale covers of different algebraically closed fields.

For the other direction we start with a given morphism $p : X \to \mathbb{P}^1$ defined over $\overline{\mathbb{Q}}$. It is étale over all but a finite number of points $S$. Our goal is then to change the morphism to one étale over all points not $\{0, 1, \infty\}$.

We first may assume that $p$ is étale except over points defined over $\mathbb{Q}$. This takes some proof.

By composing with an automorphism of $\mathbb{P}^1$, if $|S| \leq 3$, then $S \subseteq \{0, 1, \infty\}$ and we are done. Otherwise we may assume there is a point $P \in S$ defined over $\mathbb{Q}$.

Then we choose integers $A, B$ so that by composing with $x^A(x-1)^B$ so $P = \frac{A}{A+B}$ in affine space, the composed function is étale over all points as well as $P$. □

**Corollary 4.** *The outer Galois representation*

$$\rho_{\mathbb{P}^1\setminus\{0,1,\infty\}} : \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \to \mathrm{Out}(\pi_1(\mathbb{P}^1_{\overline{\mathbb{Q}}} \setminus \{0,1,\infty\})$$

*has trivial kernel.*

*Proof.* We only give the sketchiest of sketches here.

If there is non-trivial kernel, some outer representation for $\mathrm{Gal}(L)$ is trivial of $L/\mathbb{Q}$. Unwinding definitions and using the correspondence between $\pi_1$ sets and finite étale covers, every finite étale cover of $\mathbb{P}^1_{\overline{\mathbb{Q}}} \setminus \{0,1,\infty\}$ comes from a lift of one from $\mathbb{P}^1_L \setminus \{0,1,\infty\}$.

By the previous theorem this implies that every integral proper normal curved defined over $\overline{Q}$ may be defined over $L$, which there are known counterexamples to. $\square$